# Ohm’s law

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect voltage, current and resistance for an ohmic component.

**SUCCESS CRITERIA - I can...**

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Move from r to **3** r away from a point charge. **Compare field magnitude** and say what you treated as true about the source charge.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** V = IR; I = V/R.
Ohmic resistor with temperature and other physical properties kept the same. Resistance can change with temperature.

**Worked reasoning:** Across a **4 ohm** resistor, a **12 V** supply gives I=V/R=**3 A**. Doubling resistance at the same voltage halves current.

**Your turn:** Draw a one-resistor circuit, label supply voltage and current, and **mark both resistor terminals**.

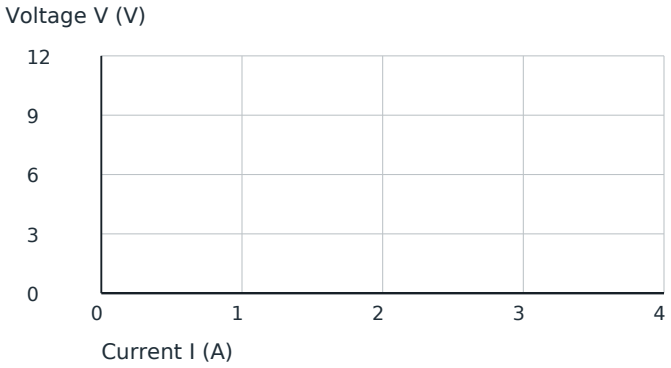



Label component values, current and voltage as needed.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

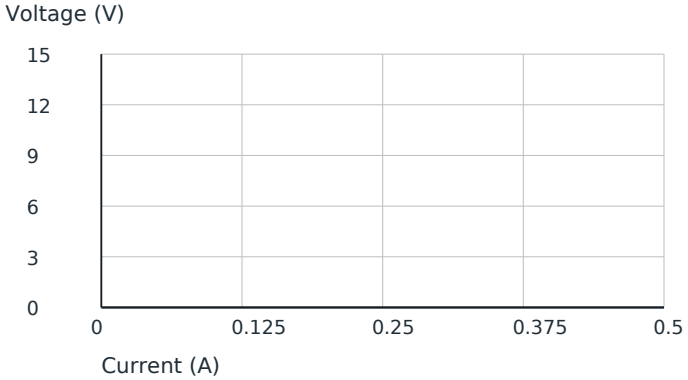
1. **Sketch a graph of V** against I for a fixed ohmic resistor.
2. What does its slope represent?




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A resistor has **12 V** across its terminals and carries **0.40 A**. **Find resistance**. If the voltage becomes **9 V** and resistance stays constant, find the new current. **Label both operating points** on the graph.




**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

An EM wave in vacuum has frequency **$6\times10^{9}$ Hz**. Use c=**$3.00\times10^{8}$ m/s**. **Find wavelength** and **name the spectral region**.
**Label a wave sketch** to **support the calculation or comparison**.

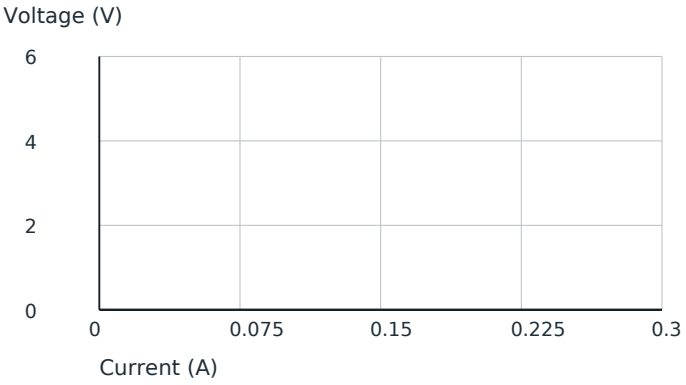



Time or position: choose and label

**Marzano check - circle:** ① ② ③

## 07 Exit check

Measured values are (I, V)=(**0.10**, **2.0**),(**0.20**, **4.0**),(**0.30**, **6.0**). **Find resistance** and **explain what supports treating the** resistance as constant.




**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

An ohmic component has **4 V** across it at **0.5 A**.
1. **Find resistance** and current at **10 V**.
2. **Explain why this extrapolation requires** the component to remain ohmic.




Label component values, current and voltage as needed.

**Marzano check - circle:** ① ② ③